$\mathcal{L}_\infty$ Ultraproducts Over Good Ultrafilters Are $\aleph$-Injective

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Status

This packet records a candidate full solution, likely valid, to Open Problem (6) of Aviles, Cabello Sanchez, Castillo, Gonzalez, and Moreno [1]: whether every ultraproduct over a countably incomplete $\aleph$-good ultrafilter is $\aleph$-injective as long as it is an $\mathcal{L}_\infty$-space.

Source Passage

> Theorem 6.10. Suppose we have $Y \subset X$ Banach spaces with $\operatorname{dens} X < \aleph$ and $t : Y \longrightarrow C(K)$ an operator of norm 1. We can apply part (2) of Theorem 6.10 to $\pi : B_{X^*} \longrightarrow B_{Y^*}$ and the mapping $f : K \longrightarrow B_{Y^*}$ given by $f(x) = t^*(\delta_x)$. We obtain a weak*-continuous function $g : K \longrightarrow B_{X^*}$ such that $\pi g = f$. Then, the formula $T(x)(k) = \|x\| g(k)(x/\|x\|)$, $x \in X$, $k \in K$, defines an extension of t of norm 1.
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> **7. Open Problems**
>
> (1) Find homological characterizations of $\aleph$-injectivity and universal $\aleph$-injectivity in ZFC. Proposition 2.2 characterizes $(2^{\aleph})^+$-injectivity; in particular, it characterizes $\aleph_2$-injectivity under CH (*a Banach space E is $\aleph_2$-injective if and only if if it is complemented in every superspace W such that W/E is a quotient of ℓ_∞*) and $\aleph^+$-injectivity under GCH (*a Banach space E is $\aleph^+$-injective if and only if it is complemented in every superspace W such that W/E is a quotient of $\ell_\infty(\aleph)$*).
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> (2) Find a characterization of universal $(1, \aleph)$-injectivity by means of intersection of families of balls.
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> (3) Is universal $\aleph$-injectivity a 3-space property?
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> (4) Is it consistent that $C(\mathbb{N}^*)$ is $\aleph_2$-injective? Recall that we have already shown that $C(\mathbb{N}^*)$ is not $(1, \aleph_2)$-injective nor $\mathfrak{c}^+$-injective.
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> (5) Are Lindenstrauss ultraproducts via countably incomplete $\aleph$-good ultrafilters universally $\aleph$-injective spaces in ZFC? (They are universally $(1, \aleph)$-injective under GCH.)
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> (6) Prove or disprove that every ultraproduct built over a countably incomplete, $\aleph$-good ultrafilter is $\aleph$-injective as long as it is a $\mathscr{L}_\infty$-space.

Transcription of the Open Problem

Open Problem (6) asks:

"Prove or disprove that every ultraproduct built over a countably incomplete, $\aleph$-good ultrafilter is $\aleph$-injective as long as it is a $\mathcal{L}_\infty$-space."

Definitions Used

Let $\aleph$ be an uncountable cardinal. A Banach space E is $\aleph$-injective if, whenever $Y \subset X$, $\operatorname{dens} X < \aleph$, and $t : Y \to E$ is an operator, there is an extension $T : X \to E$. It is $(\lambda, \aleph)$-injective if T can always be chosen with $\|T\| \le \lambda \|t\|$.

Following the source paper, an ultrafilter $\mathcal{U}$ is countably incomplete if there are $I_n \in \mathcal{U}$ with $I_{n+1} \subset I_n$ and $\bigcap_n I_n = \varnothing$. It is $\aleph$-good if every monotone map from the finite subsets of any set of cardinality $< \aleph$ into $\mathcal{U}$ has a multiplicative refinement.

A Banach space E is an $\mathcal{L}_{\infty,\lambda}$-space if every finite-dimensional subspace of E is contained in a finite-dimensional subspace whose Banach–Mazur distance to some ℓ_∞^n is at most λ. An $\mathcal{L}_\infty$-space means $\mathcal{L}_{\infty,\lambda}$ for some finite λ.

Proof Intuition

The source theorem proves the isometric case by turning a family of fewer than $\aleph$ balls into fewer than $\aleph$ internal sets and applying good-ultrafilter saturation. For a general $\mathcal{L}_\infty$ ultraproduct, one should not ask for exact ball intersections. The right finite-level input is instead the standard local extension property of $\mathcal{L}_{\infty,\lambda}$-spaces: finite-dimensional extension problems can be solved with constant λ.

The good ultrafilter then upgrades finite solvability to solvability for all constraints indexed by a dense set of size $< \aleph$. We encode a candidate extension as a function from a fixed rational dense subspace into the ultraproduct. Linearity, agreement with the original operator, and the λ-norm bound are all closed finite constraints. Every finite subcollection is solved by the finite-dimensional extension lemma, so saturation realizes the whole type.

Theorem

Theorem 1. *Let $\mathcal{U}$ be a countably incomplete $\aleph$-good ultrafilter, and let*

$$E = [E_i]_{\mathcal{U}}$$

be an ultraproduct of Banach spaces. If E is an $\mathcal{L}_{\infty,\lambda}$-space, then E is $(\lambda, \aleph)$-injective. Consequently, if E is an $\mathcal{L}_\infty$-space, then E is $\aleph$-injective.

Lemma 1 (Finite-dimensional extension)**.** *Let E be an $\mathcal{L}_{\infty,\lambda}$-space. If $A \subset B$ are finite-dimensional Banach spaces and $u : A \to E$ is an operator, then u has an extension $\widetilde{u} : B \to E$ with $\|\widetilde{u}\| \le \lambda \|u\|$.*

Proof. Let $F = u(A)$. By the $\mathcal{L}_{\infty,\lambda}$ property, F is contained in a finite-dimensional subspace $G \subset E$ and there is an isomorphism $S : G \to \ell_\infty^n$ such that $\|S\| \|S^{-1}\| \le \lambda$. Since ℓ_∞^n is 1-injective, $Su : A \to \ell_\infty^n$ extends to an operator $v : B \to \ell_\infty^n$ with $\|v\| = \|Su\|$. Then $S^{-1}v : B \to G \subset E$ extends u, and

$$\|S^{-1}v\| \le \|S^{-1}\| \|S\| \|u\| \le \lambda \|u\| .$$

□

Proof of the Theorem

We use the saturation fact stated in the proof of Theorem 5.1 of [1]: if $\mathcal{U}$ is countably incomplete and $\aleph$-good, then any family of fewer than $\aleph$ internal subsets of a set-theoretic ultraproduct with the finite intersection property has nonempty intersection.

Let $Y \subset X$ be Banach spaces with $\operatorname{dens} X < \aleph$, and let $t : Y \to E$ be an operator. The case $t = 0$ is trivial, so put $M = \|t\| > 0$. Choose a rational vector subspace $D \subset X$ which is dense in X, has cardinality $< \aleph$, and such that

$$D_Y := D \cap Y$$

is dense in Y. For each $y \in D_Y$, fix a representative $(t_i(y))_{i\in I}$ of $t(y) \in [E_i]_{\mathcal{U}}$.

Let S_i be the set of all functions $\varphi_i : D \to E_i$. An element $\langle \varphi_i \rangle_{\mathcal{U}}$ of the set-theoretic ultraproduct $\langle S_i \rangle_{\mathcal{U}}$ determines a candidate map

$$T_0 : D \to E, \qquad T_0(x) = [(\varphi_i(x))]_{\mathcal{U}}.$$

We now impose fewer than $\aleph$ internal constraints on $\langle \varphi_i \rangle_{\mathcal{U}}$.

First, for $x, x' \in D$, $q \in \mathbb{Q}$, and $m \in \mathbb{N}$, impose the approximate linearity constraints

$$\left\|\varphi_i(x + x') - \varphi_i(x) - \varphi_i(x')\right\| \leq 1/m, \qquad \|\varphi_i(qx) - q\varphi_i(x)\| \leq 1/m.$$

Second, for $y \in D_Y$ and $m \in \mathbb{N}$, impose the agreement constraint

$$\|\varphi_i(y) - t_i(y)\| \leq 1/m.$$

Finally, for every finite tuple $x_1, \ldots, x_n \in D$, rational scalars $q_1, \ldots, q_n$, and $m \in \mathbb{N}$, impose the norm constraint

$$\left\|\sum_{j=1}^{n} q_j \varphi_i(x_j)\right\| \leq \lambda M \left\|\sum_{j=1}^{n} q_j x_j\right\| + 1/m.$$

Each displayed condition defines an internal subset of $\langle S_i \rangle_{\mathcal{U}}$, and the total number of such constraints is $< \aleph$.

It remains to check finite satisfiability. Take finitely many of the above constraints. Only finitely many vectors of D and D_Y occur. Let

$$B = \operatorname{span}\{\text{all occurring vectors of } D\}, \qquad A = B \cap Y.$$

Since B is finite-dimensional, Lemma 1 extends $t|_A : A \to E$ to an operator $u : B \to E$ with $\|u\| \leq \lambda M$. Choose representatives $(u_i(x))_{i\in I}$ for the finitely many values $u(x)$ attached to vectors that occur in the chosen constraints, and define φ_i on those relevant vectors by these representatives, extending it arbitrarily to all of D. Since the required equalities and norm bounds hold in the ultraproduct, the set of i's on which the finite list of approximate coordinate inequalities holds belongs to $\mathcal{U}$. Hence the corresponding finite intersection of internal sets is nonempty.

By the good-ultrafilter saturation theorem, there is $\langle \varphi_i \rangle_{\mathcal{U}} \in \langle S_i \rangle_{\mathcal{U}}$ satisfying all constraints. The associated map $T_0 : D \to E$ is exactly $\mathbb{Q}$-linear, agrees with t on D_Y, and satisfies

$$\|T_0(x)\| \leq \lambda M \, \|x\| \qquad (x \in D).$$

Thus T_0 extends uniquely by continuity to a bounded linear operator

$$T : X \to E$$

with $\|T\| \leq \lambda \, \|t\|$. Since D_Y is dense in Y and $T_0 = t$ on D_Y, the extension satisfies $T|_Y = t$.

Therefore E is $(\lambda, \aleph)$-injective. If E is merely an $\mathcal{L}_\infty$-space, then E is $\mathcal{L}_{\infty,\lambda}$ for some finite λ, so the preceding paragraph gives $\aleph$-injectivity. This answers Open Problem (6) affirmatively. $\square$

Verification Notes

No numerical verification is involved. The finite-dimensional extension lemma is the standard $\mathcal{L}_{\infty,\lambda}$ argument through ℓ_∞^n. The only set-theoretic input is precisely the internal-set saturation property quoted and used in the source paper's proof of Theorem 5.1.

The result does not address Open Problem (5), which asks for universal $\aleph$-injectivity of Lindenstrauss ultraproducts in ZFC. The proof here requires $\operatorname{dens} X < \aleph$, exactly as in the definition of ordinary $\aleph$-injectivity.

Novelty and Literature Check

The local run indexes were searched for arXiv:1406.6733, "aleph-injective", "aleph-good", "good ultrafilter", "countably incomplete", "$\mathcal{L}_\infty$ ultraproduct", and related phrases. No existing packet or attempt covers Open Problem (6).

External web searches on 2026-06-19 for exact phrases around $\aleph$-good ultrafilters, $\mathcal{L}_\infty$ ultraproducts, and $\aleph$-injectivity found the source paper [1] and the earlier separably-injective paper [2], but no later paper explicitly resolving this higher-cardinal problem.

Human review should focus on the saturation coding in the proof: the ambient set-theoretic ultraproduct is built from the sets S_i of all functions $D \to E_i$, and each linearity, agreement, or norm bound is imposed as an internal coordinatewise condition with $1/m$ tolerance.

References

[1] A. Aviles, F. Cabello Sanchez, J. M. F. Castillo, M. Gonzalez, and Y. Moreno, *$\aleph$-injective Banach spaces and $\aleph$-projective compacta*, arXiv:1406.6733.

[2] A. Aviles, F. Cabello Sanchez, J. M. F. Castillo, M. Gonzalez, and Y. Moreno, *On separably injective Banach spaces and Corrigendum to "On separably injective Banach spaces"*, arXiv:1103.6064.

[3] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces I and II*, Classics in Mathematics, Springer, 1996.